

Saturn AL-31

The **Saturn AL-31** is a family of military turbofan engines, developed by the Lyulka, now NPO Saturn, in the Soviet Union/Russia, originally for the Sukhoi Su-27 air superiority fighter. It produces a thrust of 28,000 lbf (123 kN) with afterburning in the AL-31F, 31,000 lbf (137 kN) in the AL-31FM (AL-35F) and 33,000 lbf (145 kN) in the AL-37FU variants. Currently it powers all Su-27 derivatives and the Chengdu J-10 multirole jet fighter which has been developed under a Russian license by China.

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Saturn AL-31 FN turbofan engine	
Type	Turbofan
National origin	USSR/Russia
Manufacturer	NPO Saturn, UMPO, NPC Saljut
Major applications	Sukhoi Su-27 Sukhoi Su-30 Sukhoi Su-35 Shenyang J-11 Chengdu J-10 Chengdu J-20 Sukhoi S-70B Okhotnik-B (UCAV)
Developed from	Lyulka/Saturn AL-21F-3A
Developed into	Saturn AL-41

Variants

https://en.wikipedia.org/wiki/Saturn_AL-31#Specifications_AL-41F-1S_(117S)

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Name	Description	Builder	Year	Thrust	Thrust vectoring	Aircraft	Status
AL-31F ^{[1][2][3][4]}	The basic engine developed to power the Su-27 fighter	Salyut, UMPO	1981	27,600 lbf (122.6 kN)	No	<u>Sukhoi Su-27</u> , <u>Shenyang J-11</u> , <u>Sukhoi Su-30MKK</u> , <u>Sukhoi Su-30 (Salyut)</u>	In service/production
AL-31F3	Improved variant for the naval version Su-33	Saturn Lyul'ka		28,200 lbf (125.57 kN)	No	<u>Sukhoi Su-33</u>	In service/production
AL-31FP	Improved variant for the Indian Su-30MKI with thrust vectoring	Salyut, <u>HAL</u>	2000	27,600 lbf (122.6 kN)	Yes	<u>Sukhoi Su-30 MKI</u> , <u>Sukhoi Su-30MKM</u>	In service/production
AL-31FN ^[5]	Improved variant for the Chengdu J-10	Salyut	2002	28,000 lbf (124.5 kN)	No	<u>Chengdu J-10</u>	In service/production
AL-31FN Series 3 ^[6]	Improved variant for the Chengdu J-10B	Salyut	2013	30,200 lbf (134.3 kN)	No	<u>Chengdu J-10</u>	In service/production ^[7]
AL-31F M1 ^[8]	Improved version for the Russian Air Force	Salyut	2007	30,300 lbf (135 kN)	Yes	<u>Sukhoi Su-27SM</u> , <u>Sukhoi Su-30</u> , <u>Sukhoi Su-34</u>	In service/production
AL-31F M2 ^[9]	Improved version for the Su-34 and Chengdu J-20	Salyut	2012	32,600 lbf (145 kN)	No	<u>Sukhoi Su-34</u> , <u>Chengdu J-20</u>	In service/production
AL-37FU	Advanced derivative for the Su-37	UMPO		32,600 lbf (145 kN)	Yes	<u>Sukhoi Su-37</u>	Experimental derivative for the Su-37
AL-41F-1S (117S) ^[10]	Advanced derivative for the Su-35	UMPO	2010	31,900 lbf (142 kN)	Yes	<u>Sukhoi Su-35</u>	In service/production
AL-41F1 (117)	Advanced derivative	UMPO	2010	33,000 lbf (147 kN)	Yes	<u>Sukhoi Su-57</u>	In service/production

2021/9/11				Saturn AL-31 - Wikipedia			
	for the Sukhoi Su-57				prototype		

The AL-31FP and AL-37FU variants have thrust vectoring. The former is used in the Su-30MKI export version of the Su-30 for India & Sukhoi Su-30MKM for Malaysia . The AL-31FP can deflect its nozzle to a maximum of $\pm 15^{\circ}$ at a rate of 30°/sec. The vectoring nozzle is used primarily in the pitch plane. The AL-31FP is built in India by HAL at the Koraput facility under a deep technology transfer agreement.

It can tolerate severely distorted air flow from the intake. In the twin-engine Su-27, left and right engines are interchangeable. The Mean Time Between Overhaul (MTBO) for the AL-31F is 1000 hours with a full-life of 3000 hours.

Salyut supplies AL-31 variants to fighters operated by the People's Republic of China. The J-10 uses the AL-31FN while the J-20 uses the AL-31FM2 as an interim engine until the domestic WS-15 is ready.^[11]

According to Saturn's Victor Mihailovich Chepkin, chief designer of the 117 and 117S engines, the Chinese WS-10 was developed with the aid of the AL-31's maintenance technical documentation.,^[12] this was recently confirmed by AVIC ^[13]

AL-41F1S (117S)

Intended to power the Su-35BM, the *izdeliye* 117S, or AL-41F1S, is derived from the AL-41F1 of the Su-57, with the key difference being the separate engine control system of the AL-41F1S.^[14] The engine produces 86.3 kN (19,400 lbf) of thrust dry, 137.3 kN (30,900 lbf) in afterburner, and 142.2 kN (32,000 lbf) in an emergency.^[15] The AL-41F1S fan diameter of 932 mm (36.7 in) is 3% greater than the AL-31's 905 mm (35.6 in). This engine has an assigned life of 4,000 hours and an MTBO of 1,000 hours.^[16] The first flight of this engine was completed in an Su-35BM on 20 February 2008.^[17] On 9 August 2010, Ufa-based company UMPO started supplying 117S engines (AL-41F1S) intended for Su-35S fighters.^[18]

AL-41F1 (117)

Intended to power the fifth-generation Sukhoi Su-57 fighter, the *izdeliye* 117, or AL-41F1, is a highly improved AL-31F derivative, with some application of technology from the Saturn AL-41F. Its development began in 2004 and has increased fan diameter, new high- and low-pressure turbines, provisions for thrust-vectoring nozzles similar to the AL-31FP, and a digital control system (FADEC) integrated into the aircraft's flight controls. According to Sukhoi director Mikhail Pogosyan, the AL-41F1 is a new fifth-generation engine built specifically for the Su-57. Though the specifics remain classified, the engine's thrust was increased by 24.5 kN (5,500 lbf) over the AL-31 while the engine weight growth was reduced by 150 kg (330 lb). The AL-41F1 produces 147 kN (33,000 lbf) of thrust in afterburner and has a dry weight of approximately 1,600 kg (3,500 lb) and thrust-to-weight ratio of 9.3:1.^{[19][20]} Like the AL-31F, the AL-41F1 has 4 low-pressure compressor (fan) and 9 high-pressure compressor stages.^[21] Mikhail Pogosyan further mentioned that the AL-41F1 meets the Russian Air Force requirements and will be installed in production Su-57 fighter which will be supplied to the Russian Air Force and prospective foreign clients.^[20]

Izdeliye 30 (Item 30)

The AL-41F1 (117S) is an interim engine meant for prototype and initial production batches of the Su-57. The definitive second stage for the aircraft is designated *izdeliye* 30 and will eventually replace the AL-41F1 after the mid-2020. The *izdeliye* 30 engine will have increased thrust and fuel efficiency as well as improved reliability and lower costs. Bench testing of the new engine will start in 2014 according to the general designer-director of the NPO Saturn Eugeny Marchuk.^[22] Test flights began in 2017 being still underway in 2020. At the time it was expected that serial deliveries of Su-57 fifth-generation fighters fitted out with the new engines could begin in 2022.^[23]

Specifications (AL-31F)

Data from Gordon,^[24] *Rosoboronexport,*^[25] *United Engine Corporation*^[26]

General characteristics

- **Type:** Two-shaft afterburning turbofan
- **Length:** 4,945 mm (194.7 in)
- **Diameter:**
 - 905 mm (35.6 in) inlet;
 - 1,280 mm (50 in) maximum external
- **Dry weight:** 1,520 kg (3,350 lb)

Components

- **Compressor:** 4 fan and 9 HP compressor stages
- **Combustors:** Annular
- **Turbine:** 2 single-staged turbines

Performance

- **Maximum thrust:**
 - 74.6 kN (16,800 lb_f) dry thrust
 - 122.6 kN (27,600 lb_f) with afterburner
- **Overall pressure ratio:** 23:1
- **Bypass ratio:** 0.56:1
- **Air mass flow:** 112 kg/s (250 lb/s)
- **Turbine inlet temperature:** 1,665 K (2,537 °F; 1,392 °C)
- **Specific fuel consumption:**
 - **Dry thrust:** 22.1 g/kN/s (0.78 lb/lbf/h)
 - **With afterburner:** 55.5 g/kN/s (1.96 lb/lbf/h)
- **Thrust-to-weight ratio:** 4.93 (dry), 8.22 (afterburning)

Specifications AL-41F-1S (117S)

Data from Rosobornexport^[27]

General characteristics

- **Type:** Two-shaft afterburning turbofan
- **Length:** 194.6 in (494.2 cm)
- **Diameter:** 36.7 in (93.2 cm) inlet
- **Dry weight:** 3,536 lb (1,604 kg)

Components

- **Compressor:** Axial, 4 stage fan, 9 stage compressor
- **Combustors:** Annular
- **Turbine:** 2 single stage turbines

Performance

- **Maximum thrust:**
 - 19,400 lbf (86.3 kN) dry
 - 30,900 lbf (137.3 kN) with afterburner
 - 32,000 lbf (142.2 kN) emergency thrust
- **Turbine inlet temperature:** 2,681 °F; 1,472 °C (1,745 K),
- **Fuel consumption:**
 - 15,020 lb/h (6,813 kg/h) dry
 - 55,050 lb/h (24,969 kg/h) with afterburner
- **Specific fuel consumption:**
 - 0.790 lb/lbf/h (22.37 g/kN/s) dry
 - 1.819 lb/lbf/h (51.53 g/kN/s) with afterburner
- **Thrust-to-weight ratio:** 5.49 (dry), 8.75 with afterburner, 9.04 with emergency thrust

See also

Comparable engines

- General Electric F110
- Pratt & Whitney F100
- Klimov RD-33
- Shenyang WS-10
- Soloviev D-30

Related lists

- List of aircraft engines

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Notes

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External links

- Russian Military Analysis (<http://warfare.be/?catid=341&linkid=2493>) - AL-31 F aircraft turbofan engine
 - AL-31F on LeteckeMotory.cz (<http://www.leteckemotory.cz/motory/al-31/>) (cs,en)
 - AL-41F1A (117S) on LeteckeMotory.cz (<http://www.leteckemotory.cz/motory/al-41/>) (cs,en)
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